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# Safety data sheet



BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.
Date / Revised: 06.12.2024 Version: 2.0
Date / Previous version: 15.12.2022 Previous version: 1.0
Product: **Na-Ethylate Crystals**
(ID no. 30036707/SDS_GEN_DE/EN)
Date of print 29.06.2025

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

# Na-Ethylate Crystals

Chemical name: sodium ethylate
INDEX-Number: 603-041-00-8
CAS Number: 141-52-6

REACH registration number: 01-2119972296-27-0000

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Chemical
Recommended use: process chemical, Raw material

For the detailed identified uses of the product see appendix of the safety data sheet.

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Division Monomers

Telephone: +49 621 60 42737
E-mail address: pss.monomers@basf.com

### 1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

| | |
|---|---|
| Flam. Sol. 1 | H228 Flammable solid. |
| Self-heat. 1 | H251 Self-heating: may catch fire. |
| Acute Tox. 4 (oral) | H302 Harmful if swallowed. |
| Skin Corr. 1B | H314 Causes severe skin burns and eye damage. |
| Eye Dam. 1 | H318 Causes serious eye damage. |

According to BASF current knowledge and application of the criteria given in Annex I of Regulation (EC) No. 1272/2008, the following classification exceeding the classification given in Regulation (EC) No 1272/2008, Annex VI, Table 3.1 is required.

Flam. Sol. 1
Self-heat. 1
Acute Tox. 4 (oral)
Skin Corr. 1A
Eye Dam. 1

For the classifications not written out in full in this section the full text can be found in section 16.

### 2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

| | |
|---|---|
| H228 | Flammable solid. |
| H251 | Self-heating: may catch fire. |
| H302 | Harmful if swallowed. |
| H314 | Causes severe skin burns and eye damage. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER or physician. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):

EUH014: Reacts violently with water.
EUH071: Corrosive to the respiratory tract.

Hazard determining component(s) for labelling: sodium ethanolate

### 2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
When finely distributed, self-ignition is possible. The product is under certain conditions capable of dust explosion. Corrodes metals in the presence of water or moisture.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Chemical nature

sodium ethanolate

| | |
|---|---|
| CAS Number: 141-52-6<br>EC-Number: 205-487-5<br>INDEX-Number: 603-041-00-8 | Flam. Sol. 1<br>Self-heat. 1<br>Acute Tox. 4 (oral)<br>Eye Dam. 1<br>Skin Corr. 1B<br>H228, H251, H314, H302<br>EUH014<br>EUH071 |
| | Differing classification according to current knowledge and the criteria given in Annex I of Regulation (EC) No. 1272/2008 |
| | Flam. Sol. 1<br>Self-heat. 1 |

Skin Corr. 1A
Eye Dam. 1
Acute Tox. 4 (oral)
EUH014
EUH071

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

### 3.2. Mixtures

Not applicable

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

Immediately remove contaminated clothing. First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position).

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200 - 300 ml water, do not induce vomiting, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: skin corrosion, Eye irritation, Further symptoms are possible

Hazards: No hazard is expected under intended use and appropriate handling.

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
dry powder, Dry sand, alcohol-resistant foam

Unsuitable extinguishing media for safety reasons:
water, carbon dioxide

### 5.2. Special hazards arising from the substance or mixture

Advice: Reacts violently with water. May release highly flammable and/or corrosive gases/vapours.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## SECTION 6: Accidental Release Measures

### 6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing. Use breathing apparatus if exposed to vapours/dust/aerosol.

### 6.2. Environmental precautions

Discharge into the environment must be avoided.

### 6.3. Methods and material for containment and cleaning up

For small amounts: Sweep/shovel up. Correctly dispose of recovered product immediately.
For large amounts: Sweep/shovel up. Correctly dispose of recovered product immediately.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Breathing must be protected when large quantities are decanted without local exhaust ventilation. Protect against moisture. Protect from air. Protect from direct sunlight.

Protection against fire and explosion:
Take precautionary measures against static discharges. Sources of ignition should be kept well clear. Fire extinguishers should be kept handy. Avoid dust formation.

### 7.2. Conditions for safe storage, including any incompatibilities

Segregate from acids and acid forming substances.

Suitable materials for containers: Low density polyethylene (LDPE), Stainless steel 1.4301 (V2), Stainless steel 1.4401, glass, High density polyethylene (HDPE), Carbon steel (Iron), Stainless steel 1.4541, Stainless steel 1.4571, Alkyd resin lacquer 441
Unsuitable materials for containers: Aluminium, Galvanized carbon steel (Zinc), Lead-plated, Paper/Fibreboard, tinned carbon steel (Tinplate)
Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.

Storage class according to TRGS 510 (originally VCI, Germany): (4.2) Pyrophoric or self-heating substances

### 7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

64-17-5: ethanol

Short Term Exposure Classification: (TRGS 900 (DE))
Category II: Substances with a resorptive effect
OEL 380 mg/m3 ; 200 ppm (TRGS 900 (DE))
Ceiling limit value/factor: 4
If the occupational exposure limit value (AGW) and the biological limit value (BGW) are complied with, there should be no risk of damage for the unborn child (see TRGS 900, Number 2.7)

PNEC

freshwater: 0,96 mg/l

marine water: 0,79 mg/l

intermittent release: 2,75 mg/l

STP: 584 mg/l

sediment (freshwater): 3,6 mg/kg

sediment (marine water): 2,9 mg/kg

soil: 0,63 mg/kg

oral (secondary poisoning): 0,38 g/kg

DNEL

worker:
Long-term exposure - systemic and local effects, Inhalation: 1 mg/m3

## 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if breathable aerosols/dust are formed. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:
Use gauntlets.
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Avoid contact with the skin, eyes and clothing. Do not breathe dust. Handle in accordance with good industrial hygiene and safety practice. Avoid inhalation of dusts.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | solid | |
| Form: | powder, crystalline | |
| Colour: | white to slightly yellow | |
| Odour: | odourless | |
| Odour threshold: | not applicable, odour not perceivable | |
| melting point (decomposition): | 260 °C<br>Literature data.<br>The substance / product decomposes. | |
| Boiling point: | (1.013,25 hPa)<br>The substance / product decomposes therefore not determined. | |
| decomposition point: | >= 260 °C<br>(1.013 hPa)<br>Literature data. | |
| Flammability: | highly flammable solid | (UN Test N.1 (ready combustible solids)) |
| Lower explosion limit: | For solids not relevant for classification and labelling. | |
| Upper explosion limit: | For solids not relevant for classification and labelling. | |
| Flash point: | not applicable, the product is a solid | |
| Auto-ignition temperature: | not applicable | |
| Thermal decomposition: | > 280 °C (DTA)<br>The indicated value is for inert gas atmosphere.<br>> 50 °C<br>Risk of spontaneous ignition when exposed to air. | |
| SADT: | > 75 °C<br>Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4) | |
| pH value: | 12,8<br>(7 g/l, 20 °C) | |
| Viscosity, kinematic: | not applicable, the product is a solid | |

Viscosity, dynamic:
Study scientifically not justified.
Solubility in water: hydrolyzes, spontaneous decomposition
Solubility (qualitative) solvent(s): alcohols
soluble
*Information on: ethanol*
*Partitioning coefficient n-octanol/water (log Kow): -0,31 (measured)*
*(25 °C)*
*Literature data.*
---
Vapour pressure: 0,0000028 hPa (calculated)
(25 °C)
Relative density:
No data available.
Density: 0,868 g/cm3
(20 °C)
Literature data.
Relative vapour density (air):
The product is a non-volatile solid.

Particle characteristics

Particle size distribution: 55,0 µm (D10, ISO 13320-1)
200,0 µm (D90, ISO 13320-1)
110,0 µm (D50, ISO 13320-1)
Particle size distribution: fine particles -

## 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

Explosion hazard: not explosive
Impact sensitivity:
Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Self-heating substances and mixtures

Self heating ability: It is a substance capable of spontaneous heating. (Volume: 2,5 cm3) (UN Test N.4 (self heating substances))

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases: (Directive 92/69/EEC, A.12)
Forms no flammable gases in the presence of water.

Corrosion to metals

| | | |
|---|---|---|
| | Corrosive effect on: - Aluminium - Corrodes metals in the presence of water or moisture. | |

**Other safety characteristics**

| | | |
|---|---|---|
| Bulk density: | approx. 500 kg/m3 (< 40 °C) | (DIN 53466) |
| pKA: | not applicable | |
| Hygroscopy: | hygroscopic | |
| : | Because of the n-octanol/water distribution coefficient (log Pow) adsorption is not to be expected. The product has not been tested. The statement has been derived from the properties of the hydrolysis products. | |
| Surface tension: | Based on chemical structure, surface activity is not to be expected. | |
| Evaporation rate: | The product is a non-volatile solid. | |

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

| | | |
|---|---|---|
| Corrosion to metals: | Corrosive effect on: Aluminium Corrodes metals in the presence of water or moisture. | |
| Formation of flammable gases: | Remarks: | Forms no flammable gases in the presence of water. |
| | Method: | Flammability (contact with water) |

### 10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions

Exothermic reaction. Reacts with water and acids. Reacts with substances which contain active hydrogen. Self heating possible in the presence of air. Accumulation of fine dust may entail the risk of a dust explosion in the presence of air.

### 10.4. Conditions to avoid

Avoid humidity. Avoid contact with air.

### 10.5. Incompatible materials

Substances to avoid:
water, acids

### 10.6. Hazardous decomposition products

Hazardous decomposition products:
sodium hydroxide, ethanol

## SECTION 11: Toxicological Information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion.

Experimental/calculated data:

LD50 rat (oral): 560 mg/kg (OECD Guideline 401)
(dermal):Due to the corrosive properties of the substance higher doses cannot be tested. Study does not need to be conducted.

*Information on: ethanol*
*Experimental/calculated data:*
*LC50 rat (by inhalation): 124,7 mg/l 4 h (BASF-Test)*
*The vapour was tested.*
---

Irritation

Assessment of irritating effects:

Corrosive! Damages skin and eyes.

Experimental/calculated data:

Skin corrosion/irritation

rabbit: Corrosive. (OECD Guideline 404)

Serious eye damage/irritation

: As the product corrodes the skin, it can be expected to have a similar effect on the eyes also.

Respiratory/Skin sensitization

Assessment of sensitization:

As the substance is corrosive, conducting sensitization studies is not feasible.

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in a test with mammals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

*Information on: ethanol*
*Assessment of mutagenicity:*
*The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in a test with mammals.*
---

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

*Information on: ethanol*
*Assessment of carcinogenicity:*
*The International Agency for Research on Cancer (IARC) has classified this substance as a Group 1 (known) human carcinogen. The whole of the information assessable provides no indication of a carcinogenic effect.*
---

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies suggest a fertility impairing effect with high doses. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:

Causes developmental effects in animals at high, maternally toxic doses. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
The available information is not sufficient for the evaluation of specific target organ toxicity.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated exposure to large quantities may affect certain organs. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. After repeated administration the prominent effect is the induction of corrosion.

Aspiration hazard

not applicable

Interactive effects

No data available.

### 11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. The product gives rise to pH shifts. The ecotoxicological effects are solely caused by the pH.

Toxicity to fish:
EC50 (96 h) 12.900 mg/l, Pimephales promelas (Fish test acute, Flow through.)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

Aquatic invertebrates:
LC50 (48 h) 5.012 mg/l, Ceriodaphnia dubia (other, static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

EC50 (24 h) 857,79 mg/l, Artemia salina (other)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

Aquatic plants:
EC50 (4 d) 275 mg/l (growth rate), Chlorella vulgaris (OECD Guideline 201, static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

EC10 (4 d) 11,5 mg/l (growth rate), Chlorella vulgaris (OECD Guideline 201, static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

EC50 (7 d) 4.432 mg/l (other), Lemna gibba (other, static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

No observed effect concentration (7 d) 280 mg/l (other), Lemna gibba (other, static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

Microorganisms/Effect on activated sludge:
Toxic limit concentration (16 h) 6.500 mg/l, Pseudomonas putida (other, aquatic)
Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Chronic toxicity to fish:
No observed effect concentration (120 h) 250 mg/l, Brachydanio rerio (OECD Guideline 212, semistatic)
No data available.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (9 d) 9,6 mg/l, Daphnia magna (Daphnia test chronic, semistatic)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. Literature data.

*Information on:sodium hydroxide*
*Assessment of aquatic toxicity:*
*Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible. There is a high probability that the product is not acutely harmful to aquatic organisms.*
*The effect strongly depends on the pH-value. The data refers to the dissociated form of the substance.*

*Information on:ethanol*
*Assessment of aquatic toxicity:*
*There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.*
---

*Information on:sodium hydroxide*
*Toxicity to fish:*
*LC50 (96 h) 125 mg/l, Gambusia affinis (other, static)*
*The product will cause changes in the pH value of the test system. The result refers to an unneutralized sample. Literature data.*

*Information on:ethanol*
*Toxicity to fish:*
*LC50 (96 h) 13.000 mg/l, Salmo gairdneri, syn. O. mykiss (Fish test acute, static)*
*The details of the toxic effect relate to the nominal concentration. Literature data.*
---

*Information on:sodium hydroxide*
*Aquatic invertebrates:*
*EC50 (48 h) 40,4 mg/l, Ceriodaphnia sp. (other, static)*

*Literature data.*

*Information on:ethanol*
*Aquatic invertebrates:*
*LC50 (48 h) 12.340 mg/l, Daphnia magna (Daphnia test acute, static)*
*The details of the toxic effect relate to the nominal concentration. Literature data.*

*(48 h) 5.012 mg/l, Ceriodaphnia dubia (Daphnia test acute)*
*The details of the toxic effect relate to the nominal concentration. Literature data.*
---

*Information on:ethanol*
*Aquatic plants:*
*EC50 (4 d) 675 mg/l (growth rate), Chlorella vulgaris (Algal growth inhibition test)*
*The details of the toxic effect relate to the nominal concentration. Literature data.*
---

*Information on:ethanol*
*Microorganisms/Effect on activated sludge:*
*Toxic limit concentration (16 h) 6.500 mg/l, Pseudomonas putida (other, aquatic)*
*The details of the toxic effect relate to the nominal concentration. Literature data.*
---

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.

Soil living organisms:
LC50 (48 h) 100 - 1000 µg/cm2, Eisenia foetida (Screening test, filter paper)

Terrestrial plants:
EC50 (6 d) 7.890 - 15.780 mg/l, terrestrial plants (Screening test)
Literature data.

Other terrestrial non-mammals:
No data available.

## 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product is unstable in water. The elimination data also refer to products of hydrolysis. The organic component of the product is biodegradable.

Elimination information:
84 % BOD of COD (20 d) (other) (aerobic, domestic sewage, non-adapted) Readily biodegradable (according to OECD criteria).
Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

*Information on:ethanol*

*Assessment biodegradation and elimination (H2O):*
*Readily biodegradable (according to OECD criteria).*
---

*Information on:ethanol*
*Elimination information:*
*89 % BOD of the ThOD (14 d) (OECD 301C; ISO 9408; 92/69/EWG, C.4-F) (aerobic, Inoculum conforming to MITI requirements (OECD 301C))*
*Literature data.*

*84 % BOD of the ThOD (20 d) (other) (aerobic, activated sludge, domestic, non-adapted)*
*Literature data.*
---

Assessment of stability in water:
In contact with water the substance will hydrolyse rapidly.

Information on Stability in Water (Hydrolysis):
In contact with water the substance will hydrolyse rapidly.

## 12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.
The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Bioaccumulation potential:
No data available.

*Information on:ethanol*
*Assessment bioaccumulation potential:*
*No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).*
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## 12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Due to the product characteristics the test is impossible.

*Information on:sodium hydroxide*
*Assessment transport between environmental compartments:*
*Volatility: The substance will not evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Adsorption to solid soil phase is not expected.Study scientifically not justified.*

*Information on:ethanol*

*Assessment transport between environmental compartments:*
*Volatility: The substance will not evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Adsorption to solid soil phase is not expected.*
---

### 12.5. Results of PBT and vPvB assessment

The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

### 12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

### 12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Adsorbable organically-bound halogen (AOX):
This product contains no organically-bound halogen.

Other ecotoxicological advice:
Do not release untreated into natural waters. Due to the pH-value of the product, neutralization is generally required before discharging sewage into treatment plants. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations. The local regulations on waste-water treatment must be followed.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Hydrolyze product with excess of water under usage of the personal protection equipment and dispose of in accordance with local authority regulations.
Obtain the consent of pollution control authorities before discharging to wastewater treatment plants.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN3095 |
| UN proper shipping name: | CORROSIVE SOLID, SELF-HEATING, N.O.S. (SODIUM ETHYLATE/SODIUM ETHANOLATE) |
| Transport hazard class(es): | 8, 4.2 |
| Packing group: | I |
| Environmental hazards: | no |
| Special precautions for user: | : E |

RID

| | |
|---|---|
| UN number or ID number: | UN3095 |
| UN proper shipping name: | CORROSIVE SOLID, SELF-HEATING, N.O.S. (SODIUM ETHYLATE/SODIUM ETHANOLATE) |
| Transport hazard class(es): | 8, 4.2 |
| Packing group: | I |
| Environmental hazards: | no |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN3095 |
| UN proper shipping name: | CORROSIVE SOLID, SELF-HEATING, N.O.S. (SODIUM ETHYLATE/SODIUM ETHANOLATE) |
| Transport hazard class(es): | 8, 4.2 |
| Packing group: | I |
| Environmental hazards: | no |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 3095 |
| UN proper shipping name: | CORROSIVE SOLID, SELF-HEATING, N.O.S. (SODIUM |

| | |
|---|---|
| | ETHYLATE/SODIUM ETHANOLATE) |
| Transport hazard class(es): | 8, 4.2 |
| Packing group: | I |
| Environmental hazards: | no<br>Marine pollutant: NO |
| Special precautions for user: | EmS: F-A; S-N |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3095 |
| UN proper shipping name: | CORROSIVE SOLID, SELF-HEATING, N.O.S. (SODIUM ETHYLATE/SODIUM ETHANOLATE) |
| Transport hazard class(es): | 8, 4.2 |
| Packing group: | I |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

**Further information**
Specific national features of transport regulations must be observed. They are to be found in the shipping documents.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 40, 75, 40

Hazardous Incident Ordinance (Germany):
List entry in regulation: 1.4.1
Classification applies for standard conditions of temperature and pressure.

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
List entry in regulation: O1
Classification applies for standard conditions of temperature and pressure.

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (1) Weakly water polluting.

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

### 15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

## SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Flam. Sol. 1
Self-heat. 1
Acute Tox. 4 (oral)
Skin Corr. 1A
Eye Dam. 1

chemical industry

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Flam. Sol. | Flammable solids |
| Self-heat. | Self-heating substances and mixtures |
| Acute Tox. | Acute toxicity |
| Skin Corr. | Skin corrosion |
| Eye Dam. | Serious eye damage |
| H228 | Flammable solid. |
| H251 | Self-heating: may catch fire. |
| H302 | Harmful if swallowed. |
| H314 | Causes severe skin burns and eye damage. |

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

## Annex: Exposure Scenarios

**Index**

**1.** Manufacture of substance, Formulation of solutions
IS; IS; ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

**2.** Formulation & (re)packing of substances and mixtures, (liquid products)
IS; IS; ERC2; PROC3, PROC8a, PROC8b, PROC9, PROC15

**3.** Manufacture of substance, (handling as solid)
IS; IS; ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

**4.** Formulation & (re)packing of substances and mixtures, (handling as solid)
IS; IS; ERC2; PROC3, PROC8a, PROC8b, PROC9, PROC15

**5.** Use in chemical synthesis, (liquid products), (handling as solid)
IS; IS; ERC4, ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

**6.** Use as an intermediate
IS; IS; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

**7.** Use in laboratories, Use as laboratory reagent/agent
PW; PW; ERC8a, ERC8b; PROC15

**********

### 1. Short title of exposure scenario

Manufacture of substance, Formulation of solutions
IS; IS; ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC1: Manufacture of the substance<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0284 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,028354 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid | |

| | |
|---|---|
| frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |

| | |
|---|---|
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |

| | |
|---|---|
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

**********

## 2. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures, (liquid products)
IS; IS; ERC2; PROC3, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC2: Formulation into mixture<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use | |

| | |
|---|---|
| suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see | |

| |
|---|
| exposure estimates) |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

**********

## 3. Short title of exposure scenario

Manufacture of substance, (handling as solid)
IS; IS; ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC1: Manufacture of the substance<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process |

| | |
|---|---|
| | without likelihood of exposure or processes with equivalent containment conditions.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,01 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,01 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance | 0,000275 Pa |

| | |
|---|---|
| during use | |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air | Effectiveness: 30 % |

| | |
|---|---|
| changes per hour) | |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in | |

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| place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |

| | |
|---|---|
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |

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| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |

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|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,9 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,9 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |

| | |
|---|---|
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |

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|---|---|
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,875 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |

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|---|---|
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,9 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,9 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid | |

| | |
|---|---|
| frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,84 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,84 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** |
|---|

| | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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## 4. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures, (handling as solid)
IS; IS; ERC2; PROC3, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC2: Formulation into mixture<br>As no environmental hazard was identified no |

| | |
|---|---|
| | environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities |

| | |
|---|---|
| | Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,9 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,9 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant | |

| | |
|---|---|
| gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,7 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,7 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. | |

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| Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,875 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,9 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,9 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,84 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,84 |
| Assessment method | Qualitative assessment |

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|---|---|
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, high dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | Solid, medium dustiness |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,5 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,5 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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## 5. Short title of exposure scenario

Use in chemical synthesis, (liquid products), (handling as solid)
IS; IS; ERC4, ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |

| | |
|---|---|
| | |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0284 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,028354 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid | |

| | |
|---|---|
| frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |

| | |
|---|---|
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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## 6. Short title of exposure scenario

Use as an intermediate
IS; IS; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6a: Use of intermediate<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0284 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,028354 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |

| | |
|---|---|
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |

| | |
|---|---|
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |

| |
|---|
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate |

| | |
|---|---|
| | Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |

| ***Exposure estimate and reference to its source*** | |
|---|---|
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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## 7. Short title of exposure scenario

Use in laboratories, Use as laboratory reagent/agent
PW; PW; ERC8a, ERC8b; PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8a: Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8b: Widespread use of reactive processing aid (no inclusion into or onto article, indoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | sodium ethanolate<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0,000275 Pa |

| | |
|---|---|
| Process temperature | 20 °C |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| ***Risk Management Measures*** | |
| Avoid skin contact. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Avoid contact with eyes. Provide specific employee training to prevent/minimize exposures. Avoid frequent and direct contact with substance. | |
| Wear suitable working clothes., Use suitable chemically resistant gloves. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v3.6, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2835 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,283543 |
| Assessment method | Qualitative assessment |
| | Worker - dermal |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates) | |

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